

|      | Internal Exam | Assignment | Project Evaluation | End Semester Examinations |
|------|---------------|------------|--------------------|---------------------------|
| CO 1 | ✓             |            |                    | ✓                         |
| CO 2 | ✓             |            |                    | ✓                         |
| CO 3 | ✓             |            |                    | ✓                         |
| CO 4 |               | ✓          |                    | ✓                         |
| CO 5 | ✓             | ✓          |                    | ✓                         |
| CO 6 | ✓             |            |                    |                           |

|                |                                                                         |                  |                   |                    |             |
|----------------|-------------------------------------------------------------------------|------------------|-------------------|--------------------|-------------|
| Programme      | BSc Statistics                                                          |                  |                   |                    |             |
| Course Code    | STA3MN209 (P)                                                           |                  |                   |                    |             |
| Course Title   | Statistical inference                                                   |                  |                   |                    |             |
| Type of Course | Minor                                                                   |                  |                   |                    |             |
| Semester       | III                                                                     |                  |                   |                    |             |
| Academic Level | 200 - 299                                                               |                  |                   |                    |             |
| Course Details | Credit                                                                  | Lecture per week | Tutorial per week | Practical per week | Total Hours |
|                | 4                                                                       | 3                | -                 | 2                  | 75          |
| Pre-requisites | Basic knowledge of random variable, probability, standard distributions |                  |                   |                    |             |

|                |                                                                                                                     |
|----------------|---------------------------------------------------------------------------------------------------------------------|
| Course Summary | Equip students with a comprehensive understanding of sampling theory and its applications in statistical inference. |
|----------------|---------------------------------------------------------------------------------------------------------------------|

### Course Outcomes (CO):

| CO                                                                                                                                                                                             | CO Statement                                                                                                                                                                               | Cognitive Level* | Knowledge Category# | Evaluation Tools used                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------|----------------------------------------------------------------------------------|
| CO1                                                                                                                                                                                            | Define parameters and statistics, and differentiate between them using examples to illustrate their roles in statistical inference.                                                        | U                | C                   | Instructor-created exams / Quiz                                                  |
| CO2                                                                                                                                                                                            | Explain what a sampling distribution is and differentiate it from a population distribution.                                                                                               | U                | F                   | Practical Assignment / Observation of Practical Skills/ Instructor-created exams |
| CO3                                                                                                                                                                                            | Describe the procedures for statistical estimation, including point estimation and interval estimation, and analyze data to help entrepreneurial decisions using critical thinking skills. | U                | C                   | Seminar Presentation / Group Tutorial Work/ Instructor-created exams             |
| CO4                                                                                                                                                                                            | Identify unbiased and efficient estimators and apply them to estimate population parameters such as the mean, proportion, and variance.                                                    | U                | C                   | Instructor-created exams / Home Assignments                                      |
| CO5                                                                                                                                                                                            | Interpret interval estimators for population mean and population proportion and critically evaluate ethical implications of statistical methods aligning with human values.                | Ap               | P                   | One Minute Reflection Writing assignments/ Instructor-created exams              |
| CO6                                                                                                                                                                                            | Perform testing of hypothesis using any software.                                                                                                                                          | Ap               | P                   | Viva Voce/ Instructor-created exams                                              |
| * - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)<br># - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M) |                                                                                                                                                                                            |                  |                     |                                                                                  |

### COURSE CONTENT

| Mo | Units | Content | Hrs | Marks |
|----|-------|---------|-----|-------|
|----|-------|---------|-----|-------|

|                  |                                                                                                                                                                                                            |                                                                                                                                                                                             |                     |             |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-------------|
| <b>dul<br/>e</b> |                                                                                                                                                                                                            |                                                                                                                                                                                             | <b>(45<br/>+30)</b> | <b>(70)</b> |
|                  | <b>SAMPLING THEORY</b>                                                                                                                                                                                     |                                                                                                                                                                                             | <b>10</b>           | <b>10</b>   |
| <b>1</b>         | <b>1</b>                                                                                                                                                                                                   | Parameter and statistic                                                                                                                                                                     | 2                   |             |
|                  | <b>2</b>                                                                                                                                                                                                   | Sampling Distribution                                                                                                                                                                       | 2                   |             |
|                  | <b>3</b>                                                                                                                                                                                                   | Principles of sampling                                                                                                                                                                      | 2                   |             |
|                  | <b>4</b>                                                                                                                                                                                                   | Sampling distribution of a statistic                                                                                                                                                        | 2                   |             |
|                  | <b>5</b>                                                                                                                                                                                                   | Central limit theorem                                                                                                                                                                       | 2                   |             |
|                  | <b>Sections from References:</b><br><br>Unit 1: 15.4 [Ref 1]<br><br>Unit 2: 15.4.1, 15.4.2 [Ref 1]<br><br>Unit 3: 15.5 [Ref 1]<br><br>Unit 4: 16.3 [Ref 1]<br><br>Unit 5: 16.3.2 [Ref 1]                   |                                                                                                                                                                                             |                     |             |
|                  | <b>THEORY OF ESTIMATION</b>                                                                                                                                                                                |                                                                                                                                                                                             | <b>11</b>           | <b>25</b>   |
| <b>2</b>         | <b>6</b>                                                                                                                                                                                                   | Statistical Estimation Procedures-Point Estimation and Interval estimation                                                                                                                  | 2                   |             |
|                  | <b>7</b>                                                                                                                                                                                                   | Point estimation- Estimator and Estimate (Definition, Concept), Unbiases Estimator and Efficient Estimator, Point Estimators of Population Mean, Population Proportion, Population Variance | 3                   |             |
|                  | <b>8</b>                                                                                                                                                                                                   | Interval estimation-Definition                                                                                                                                                              | 3                   |             |
|                  | <b>9</b>                                                                                                                                                                                                   | Size of the random sample for specified precision                                                                                                                                           | 3                   |             |
|                  | <b>Sections from References:</b><br><br>Unit 6: 16.2 [Ref 1]<br><br>Unit 7: 16.2.1, 16.2.2, 1.62.3, 16.2.4, 16.2.5 [Ref 1]<br><br>Unit 8: 16.4, 16.4.1, 16.4.2, 16.4.3 [Ref 1]<br><br>Unit 9: 16.5 [Ref 1] |                                                                                                                                                                                             |                     |             |
|                  | <b>HYPOTHESIS TESTING</b>                                                                                                                                                                                  |                                                                                                                                                                                             | <b>12</b>           | <b>20</b>   |
| <b>3</b>         | <b>10</b>                                                                                                                                                                                                  | Testing of hypothesis, simple and composite hypothesis, null and alternate hypothesis                                                                                                       | 2                   |             |

|          |                                                                                                                                                                                                                                                                                              |                                                           |           |           |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------|-----------|
|          | <b>11</b>                                                                                                                                                                                                                                                                                    | Types of errors, Size and power of tests, critical region | 2         |           |
|          | <b>12</b>                                                                                                                                                                                                                                                                                    | One tailed and two tailed tests                           | 1         |           |
|          | <b>13</b>                                                                                                                                                                                                                                                                                    | P- value or probability value of test statistic           | 1         |           |
|          | <b>14</b>                                                                                                                                                                                                                                                                                    | Large sample tests                                        | 2         |           |
|          | <b>15</b>                                                                                                                                                                                                                                                                                    | Test for single proportion                                | 2         |           |
|          | <b>16</b>                                                                                                                                                                                                                                                                                    | Test for single mean                                      | 2         |           |
|          | <b>Sections from References:</b><br><br>Unit 10: 16.6, 16.6.1, 16.6.3, 16.6.4 [Ref 1]<br><br>Unit 11: 16.6.5, 16.6.6, 16.6.7 [Ref 1]<br><br>Unit 12: 16.6.8 [Ref 1]<br><br>Unit 13: 16.6.10 [Ref 1]<br><br>Unit 14: 17 [Ref 1]<br><br>Unit 15: 17.2.1 [Ref 1]<br><br>Unit 16: 17.3.1 [Ref 1] |                                                           |           |           |
| <b>4</b> | <b>CHI SQUARE TEST</b>                                                                                                                                                                                                                                                                       |                                                           | <b>12</b> | <b>15</b> |
|          | <b>17</b>                                                                                                                                                                                                                                                                                    | Probability density function of Chi- square distribution  | 1         |           |
|          | <b>18</b>                                                                                                                                                                                                                                                                                    | Applications of Chi square distribution                   | 2         |           |
|          | <b>19</b>                                                                                                                                                                                                                                                                                    | Chi square test of goodness of fit                        | 3         |           |
|          | <b>20</b>                                                                                                                                                                                                                                                                                    | Conditions for the validity for Chi square test           | 2         |           |
|          | <b>21</b>                                                                                                                                                                                                                                                                                    | Chi square test for independence of attributes            | 3         |           |
|          | <b>22</b>                                                                                                                                                                                                                                                                                    | Degrees of freedom                                        | 1         |           |
|          | <b>Sections from References:</b><br><br>Unit 17: 18.2.1 [Ref 1]<br><br>Unit 18: 18.3 [Ref 1]<br><br>Unit 19: 18.4 [Ref 1]<br><br>Unit 20: 18.5 [Ref 1]<br><br>Unit 21: 18.6 [Ref 1]                                                                                                          |                                                           |           |           |
|          |                                                                                                                                                                                                                                                                                              |                                                           |           |           |



|      |   |   |   |   |   |   |   |   |   |   |   |   |
|------|---|---|---|---|---|---|---|---|---|---|---|---|
| 3    |   |   |   |   |   |   |   |   |   |   |   |   |
| CO 4 | 1 | - | — | - | - | 3 | - | - | - | - | - | 3 |
| CO 5 | - | 3 | 2 | 2 | - | - | - | - | 2 | - | - | - |
| CO 6 | - | - | 3 | - | - | 3 | - | - | - | 2 | 3 | - |

**Correlation Levels:**

| Level | Correlation        |
|-------|--------------------|
| -     | Nil                |
| 1     | Slightly / Low     |
| 2     | Moderate / Medium  |
| 3     | Substantial / High |

**Assessment Rubrics:**

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

**Mapping of COs to Assessment Rubrics :**

|      | Internal Exam | Assignment | Project Evaluation | End Semester Examinations |
|------|---------------|------------|--------------------|---------------------------|
| CO 1 | ✓             |            |                    | ✓                         |
| CO 2 | ✓             | ✓          |                    | ✓                         |
| CO 3 | ✓             |            |                    | ✓                         |